

MUHAMMAD HAIDER AKBAR

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EDUCATION

Bachelors Degree in Data Science

National University of Computing and Emerging Sciences | FAST

Aug 2022 - July 2026

Islamabad, Pakistan

PROFESSIONAL EXPERIENCE

Software Developer

Epistemy (Remote)

Sep 2025 - Nov 2025

London, UK

- Architected and deployed a full-stack AI tutoring platform in **TypeScript** (Next.js + NestJS) with **Supabase** and **S3**.
- Automated video ingestion and transcription via **Deepgram** ASR, **BullMQ** background jobs and **LangGraph** chunking.
- Built **RAG pipelines** over vectorized transcripts to generate summaries, quizzes and personalized feedback.
- Owned **CI/CD** monorepo deployment to **Vercel** with **LangSmith** tracing and JWT-based role access control.

AI Engineer

DMlytics

Feb 2025 - May 2025

Islamabad, Pakistan

- Built an orthomosaic pipeline on **GCP** processing **300GB+** of aerial imagery end-to-end using **GDAL**.
- Implemented EXIF/RTK georeferencing from GPS metadata and **GSD mapping** with **Rasterio** for accuracy.
- Applied **OpenCV** image cleanup and alignment to improve tile quality before mosaicking and GeoTIFF export.
- Automated Pix4D-ready **GeoTIFF** generation with compression, tiling and cloud-sync for GIS integration.

AI Intern

FRACK Technologies

Sep 2024 - Oct 2024

Islamabad, Pakistan

- Built a FER indexing pipeline over images and annotations creating stratified train/val parquet splits for 8 classes.
- Fine-tuned **ResNet50** and EfficientNet in **PyTorch** for emotion classification and valence-arousal regression.
- Configured AdamW, LR scheduling, mixed precision and early stopping evaluated on accuracy, F1 and RMSE.

PROJECTS

JobSwipe **Flask** | **Next.js** | **Supabase** | **FAISS** | **Redis** | **LLM**

- Built a resume analysis platform with Flask API, Next.js frontend and Supabase storage.
- Implemented NLP pipelines with Sentence Transformers and FAISS to match candidates to roles.
- Powered a Reels-style swipe UI with auto-generated tailored applications for right-swiped jobs.
- Optimized LLM parsing via Redis caching and daemon threads to accelerate background processing.

AI Tutoring Assistant 🔄 **FastAPI** | **React** | **LangChain** | **RAG** | **Chroma** | **LangSmith**

- Built AI tutoring MVP with FastAPI backend, React frontend and Deepgram ASR transcription pipelines.
- Implemented semantic chunking with text-embedding-3-large and Chroma for concept retrieval tracking.
- Designed RAG-based personalized feedback and quiz generation with PDF export via Playwright.
- Monitored AI workflows end-to-end with LangSmith tracing across sessions and retrieval chains.

MondAI 🔄 **LangChain** | **Anthropic** | **GraphQL** | **MCP** | **n8n**

- Built Monday.com workflow automation with n8n, LangChain agents and Claude Sonnet for NL parsing.
- Designed memory-buffered context system with metadata-driven GraphQL calls to MCP endpoints.
- Automated board actions including item creation, updates and status queries conversationally.
- Optimized prompt engineering for error handling, batch ops and dynamic column mapping.

RhythmRabbit 🔄 **PySpark** | **Kafka** | **Azure** | **MongoDB** | **Librosa** | **Flask**

- Built distributed music recommender on Azure VMs processing 80GB+ of audio and metadata.
- Implemented Librosa ETL for feature extraction and KNN similarity search in PySpark ML.
- Designed Kafka pipelines for real-time user interaction capture and dynamic playlist updates.
- Achieved sub-200ms recommendations via indexed MongoDB retrieval under concurrent load.

HONORS & AWARDS

- Dean's List of Honors - FAST NUCES Islamabad
- Winner Hackathon - HackFest FCS Charity Gala Islamabad
- Runner Up Hackathon - Data Quest DSS CUI Tech Fest FA'25

TECHNICAL SKILLS

Languages: Python | SQL | TypeScript | JavaScript | C++

AI / ML: PyTorch | TensorFlow | Scikit-learn | HuggingFace | LangChain | RAG | Pandas | NumPy

Data & Cloud: Spark | Kafka | AWS | GCP | Docker | CI/CD | Linux

Web & Databases: FastAPI | Flask | Next.js | PostgreSQL | MongoDB | Redis | Supabase